

FUNDAMENTALS OF MACHINE LEARNING

PARAMETRIC METHODS

CSCI3320

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Introduction to Machine Learning

From Bayesian to Parametric Methods

In Chapter 3, we wrote posterior probability for class C_i

$$P(C_i|x) = \frac{p(x|C_i)P(C_i)}{p(x)} = \frac{p(x|C_i)P(C_i)}{\sum_{k=1}^K p(x|C_k)P(C_k)}$$

Using discriminant function

$$g_i(x) = p(x|C_i)P(C_i) \quad \text{or} \quad g_i(x) = \log p(x|C_i) + \log P(C_i)$$

We know how to get the prior $P(C_i)$,
but **we have problem** in getting $p(x|C_i)$

In parametric methods, we assume $p(x|C_i)$ follows some probability distributions, so we can use the probability density function of x to represent $p(x|C_i)$. The only question is how to “*estimate*” parameters of that distribution.

Outline

- Previously, we studied how to make optimal decision when uncertainty is modeled by probability (Bayes' rule)
- Focus of **SINGLE** dimension (or feature) first
- Now we learn how to *estimate these probabilities* from the training set
- **Parametric approach**
 - ▣ Classification
 - ▣ Regression
- Introduce **bias/variance dilemma**
- Introduce **model selection methods**

Parametric Estimation

- Assume samples are drawn from some distribution (e.g., Gaussian, Bernoulli,...etc)
- Once we know the parameters (e.g., mean, variance), we know the whole distribution
- **IID samples:** $X = \{x^t\}_{t=1}^N$ where $x^t \sim p(x|\theta)$
- **Parametric estimation:**
Assume a form for $p(x|\theta)$ and estimate θ , its sufficient statistics, using X
e.g., $N(\mu, \sigma^2)$ where $\theta = \{\mu, \sigma^2\}$
- Like to estimate the parameter θ **as likely as possible**

Maximum Likelihood Estimation

- **Likelihood** of θ given the sample X (due to iid)

$$l(\theta|X) \equiv p(X|\theta) = \prod_{t=1}^N p(x^t|\theta)$$

find θ which maximizes the likelihood function $l()$

*Given example of flipping a coin
N times to estimate p*

- **Log likelihood** (sometimes to reduce computation)

$$\mathcal{L}(\theta|X) \equiv \log l(\theta|X) = \sum_{t=1}^N \log p(x^t|\theta)$$

- **Maximum likelihood estimator (MLE)**

$$\theta^* = \operatorname{argmax}_{\theta} \mathcal{L}(\theta|X)$$

Example: Bernoulli

- **Bernoulli:** Two states, failure/success, N iid x^t

$$P(x) = p^x (1 - p)^{1-x} \quad x \in \{0, 1\}$$

$$\mathcal{L}(p|\mathcal{X}) = \log \prod_{t=1}^N p^{(x^t)} (1 - p)^{(1-x^t)}$$

$$= \sum_t x^t \log p + \left(N - \sum_t x^t \right) \log(1 - p)$$

- To maximize $\mathcal{L}(p|\mathcal{X})$, take derivative with p and equate to 0
- MLE:

$$\hat{p} = \frac{\sum_t x^t}{N}$$

How?

Example: Bernoulli (derivation)

$$\begin{aligned}\mathcal{L}(p|\mathcal{X}) &= \log \prod_{t=1}^N p^{(x^t)} (1-p)^{(1-x^t)} \\ &= \sum_t x^t \log p + \left(N - \sum_t x^t \right) \log(1-p)\end{aligned}$$

$$\frac{d\mathcal{L}(p|X)}{dp} = \frac{1}{p} \sum_t x^t - \left(N - \sum_t x^t \right) \frac{1}{1-p} = 0$$

$$p \left(N - \sum_t x^t \right) = (1-p) \left(\sum_t x^t \right)$$

$$pN = \sum_t x^t$$

$$p = \frac{1}{N} \sum_t x^t$$

*Explain how to get the MLE of
Bernoulli parameter
with N points*

Application

In Chapter 3, we wrote posterior probability for class C_i

$$P(C_i|x) = \frac{p(x|C_i)P(C_i)}{p(x)} = \frac{p(x|C_i)P(C_i)}{\sum_{k=1}^K p(x|C_k)P(C_k)}$$

Using discriminant function

$$g_i(x) = p(x|C_i)P(C_i) \quad \text{or} \quad g_i(x) = \log p(x|C_i) + \log P(C_i)$$

$$g_i(x) = \log (\hat{p}^x (1 - \hat{p})^{1-x}) + \log P(C_i)$$

Now we remove the limitation of Bayesian method

Application

Values	Class #
--------	---------

1	1
---	---

0	1
---	---

0	2
---	---

1	1
---	---

1	2
---	---

.....	
-------	--

1	
---	--

	?
--	---

- N training data
- Value indicates whether the person is high risk (1) or low risk (0)
- Person can belong to class 1 or class 2

Example: Multinomial

- Outcome of the event is one of K mutually exclusive and exhaustive states (or classes) with p_i with $\sum_{i=1}^K p_i = 1$.
- Let $x_1, x_2, \dots, x_K$ are the indicator variables
- x_i is 1 if the outcome is state (class) i and 0 otherwise
- We have $P(x_1, x_2, \dots, x_K) = \prod_{i=1}^K p_i^{x_i}$
- N independent experiments:

$$\mathcal{X} = \{\mathbf{x}^t\}_{t=1}^N \quad \text{Note: } \mathbf{x}^t \text{ is a vector !!!}$$

$$x_i^t = \begin{cases} 1 & \text{if experiment } t \text{ chooses state } i \\ 0 & \text{otherwise} \end{cases} \quad \text{with } \sum_i x_i^t = 1.$$

- MLE:** $\hat{p}_i = \frac{\sum_t x_i^t}{N}$

Application

Values	Class #
--------	---------

1	1
---	---

0	1
---	---

0	2
---	---

2	1
---	---

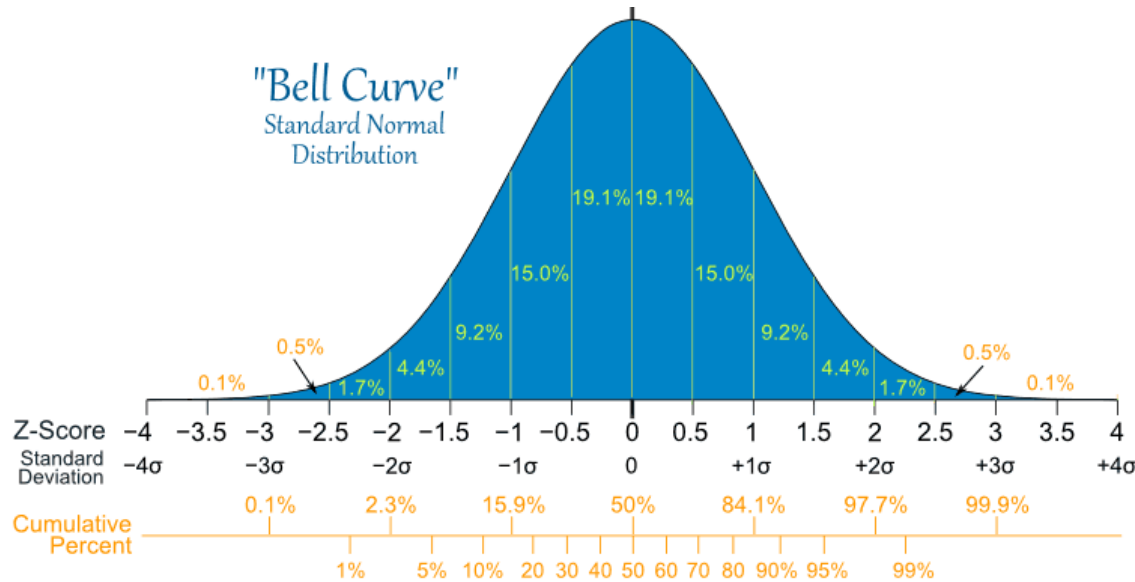
2	2
---	---

.....	
-------	--

2	?
---	---

- N training data
- Value indicates whether the person is high risk (1), low risk (0), or neutral (2)
- Person can belong to class 1 or class 2

Gaussian (Normal) Distribution



□ $p(x) : \mathcal{N}(\mu, \sigma^2)$

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right]$$

□ MLE for μ and σ^2 :

Log likelihood function

$$\mathcal{L}(\mu, \sigma | \mathcal{X}) = -\frac{N}{2} \log(2\pi) - N \log \sigma - \frac{\sum_t (x^t - \mu)^2}{2\sigma^2}$$

Show this in class

$$m = \frac{\sum_t x^t}{N}$$

$$s^2 = \frac{\sum_t (x^t - m)^2}{N}$$

Gaussian (Normal) Distribution

Log likelihood function: $\mathcal{L}(\mu, \sigma | \mathcal{X}) = -\frac{N}{2} \log(2\pi) - N \log \sigma - \frac{\sum_t (x^t - \mu)^2}{2\sigma^2}$

$$\frac{d\mathcal{L}(\mu, \sigma | \mathcal{X})}{d\mu} = \frac{1}{2\sigma^2} (2) \sum_t (x^t - \mu) = 0$$

$$N\mu = \sum_t x^t$$

$$\mu = \frac{\sum_t x^t}{N}$$

$$m = \frac{\sum_t x^t}{N}$$

$$s^2 = \frac{\sum_t (x^t - m)^2}{N}$$

$$\frac{d\mathcal{L}(\mu, \sigma | \mathcal{X})}{d\sigma} = -\frac{N}{\sigma} + 2 \frac{\sum_t (x^t - \mu)^2}{2\sigma^3} = 0$$

$$N\sigma^2 = \sum_t (x^t - \mu)^2$$

$$\sigma^2 = \frac{\sum_t (x^t - \mu)^2}{N}$$

Parametric Classification

In Chapter 3, we wrote posterior probability for class C_i

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Using discriminant function

$$g_i(x) = p(x|C_i)P(C_i) \quad \text{or} \quad g_i(x) = \log p(x|C_i) + \log P(C_i)$$

If we can assume that $p(x|C_i)$ are Gaussian

$$p(x|C_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right] \quad \sigma_1, \mu_i, P(C_i) = ?$$

$$g_i(x) = -\frac{1}{2} \log 2\pi - \log \sigma_i - \frac{(x - \mu_i)^2}{2\sigma_i^2} + \log P(C_i)$$

Classification of K cars

- Given the sample $\mathcal{X} = \{\mathbf{x}^t, \mathbf{r}^t\}_{t=1}^N$

$$\mathbf{x} \in \mathfrak{R} \quad \mathbf{r} \in \{0, 1\}^K \text{ such that } r_i^t = \begin{cases} 1 & \text{if } \mathbf{x}^t \in C_i \\ 0 & \text{if } \mathbf{x}^t \in C_k, k \neq i \end{cases}$$

- Want posterior probability $P(C_i | \mathbf{x})$, use Bayes Rule
- ML estimates** are

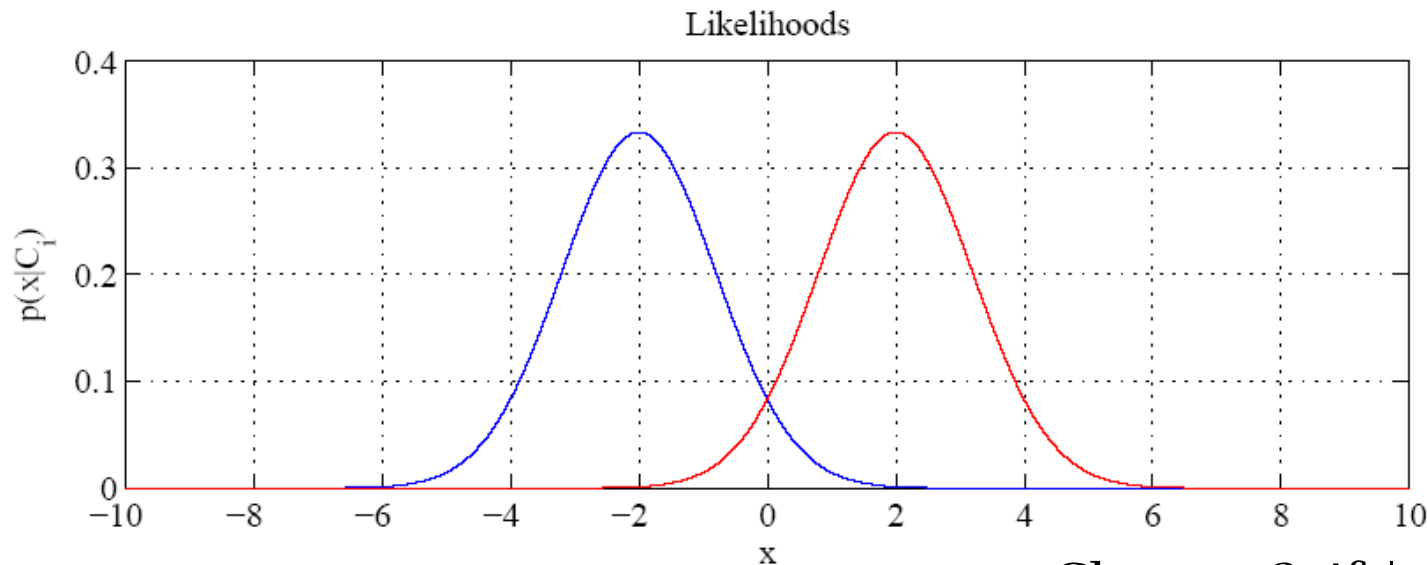
$$\hat{P}(C_i) = \frac{\sum_t r_i^t}{N} \quad m_i = \frac{\sum_t \mathbf{x}^t r_i^t}{\sum_t r_i^t} \quad s_i^2 = \frac{\sum_t (\mathbf{x}^t - m_i)^2 r_i^t}{\sum_t r_i^t}$$

- Discriminant

$$g_i(\mathbf{x}) = -\frac{1}{2} \log 2\pi - \log s_i - \frac{(\mathbf{x} - m_i)^2}{2s_i^2} + \log \hat{P}(C_i)$$

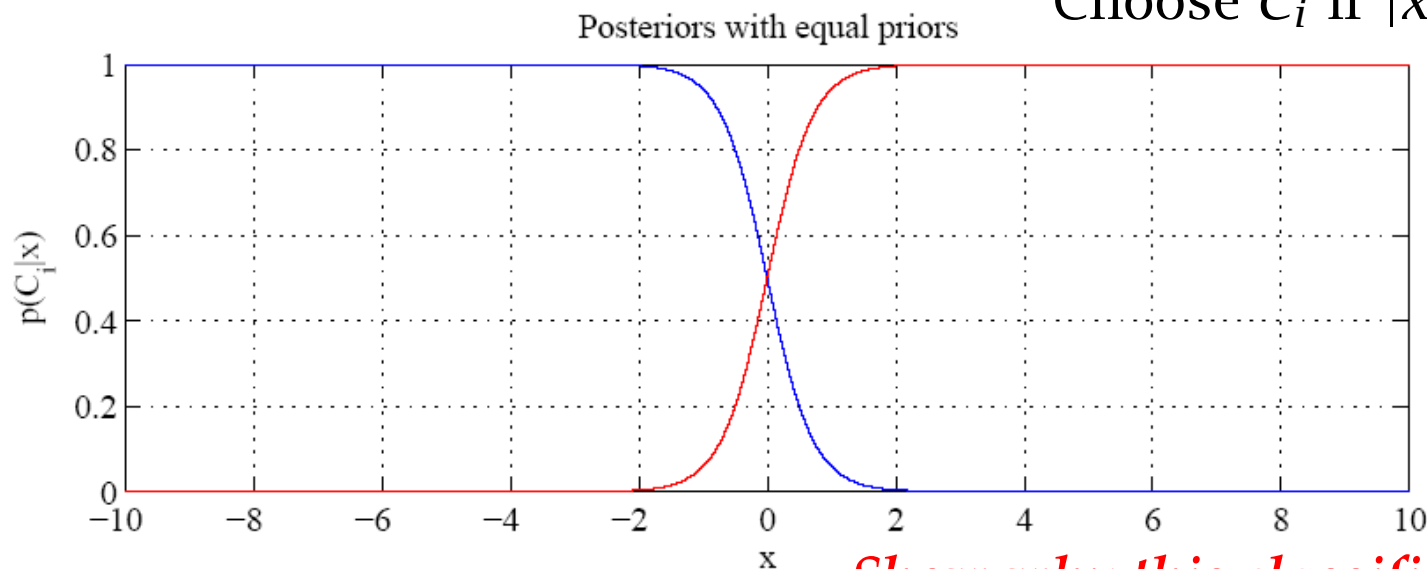
- If prior probability and variances for **ALL CLASSES ARE THE SAME:** $g_i(\mathbf{x}) = -(\mathbf{x} - m_i)^2$ *What is the physical meaning?*

When prior ($P(C_i)$) and variance (s_i^2) are equal



$$g_i(x) = -(x - m_i)^2$$

Choose C_i if $|x - m_i| = \min_k |x - m_k|$



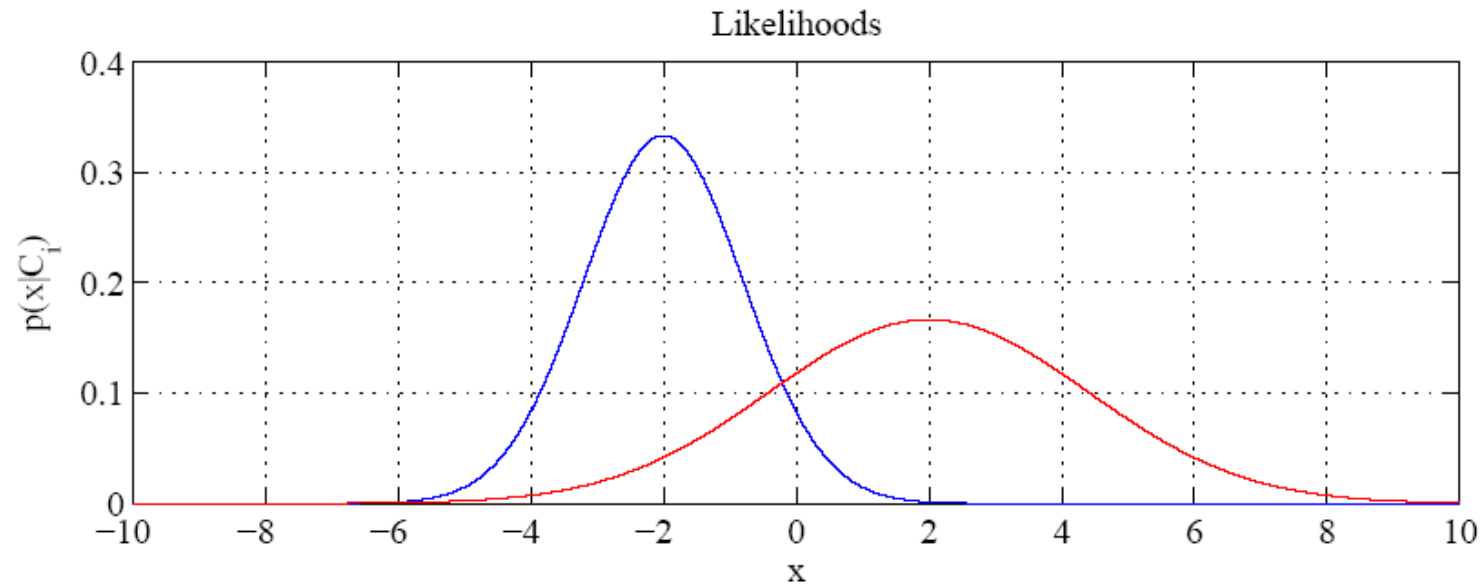
Single boundary at
halfway between means,
or when $g_1(x) = g_2(x)$

Show derivation

$$x = \frac{(m_1 + m_2)}{2}$$

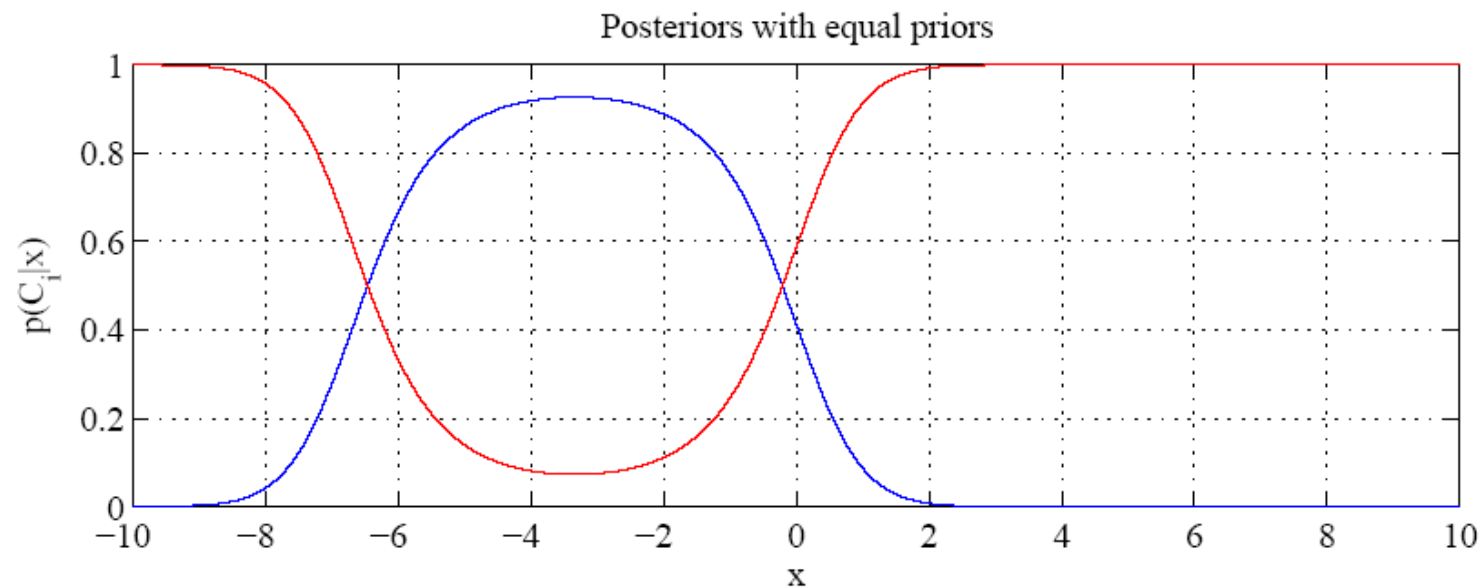
Show why this classification is intuitive 16

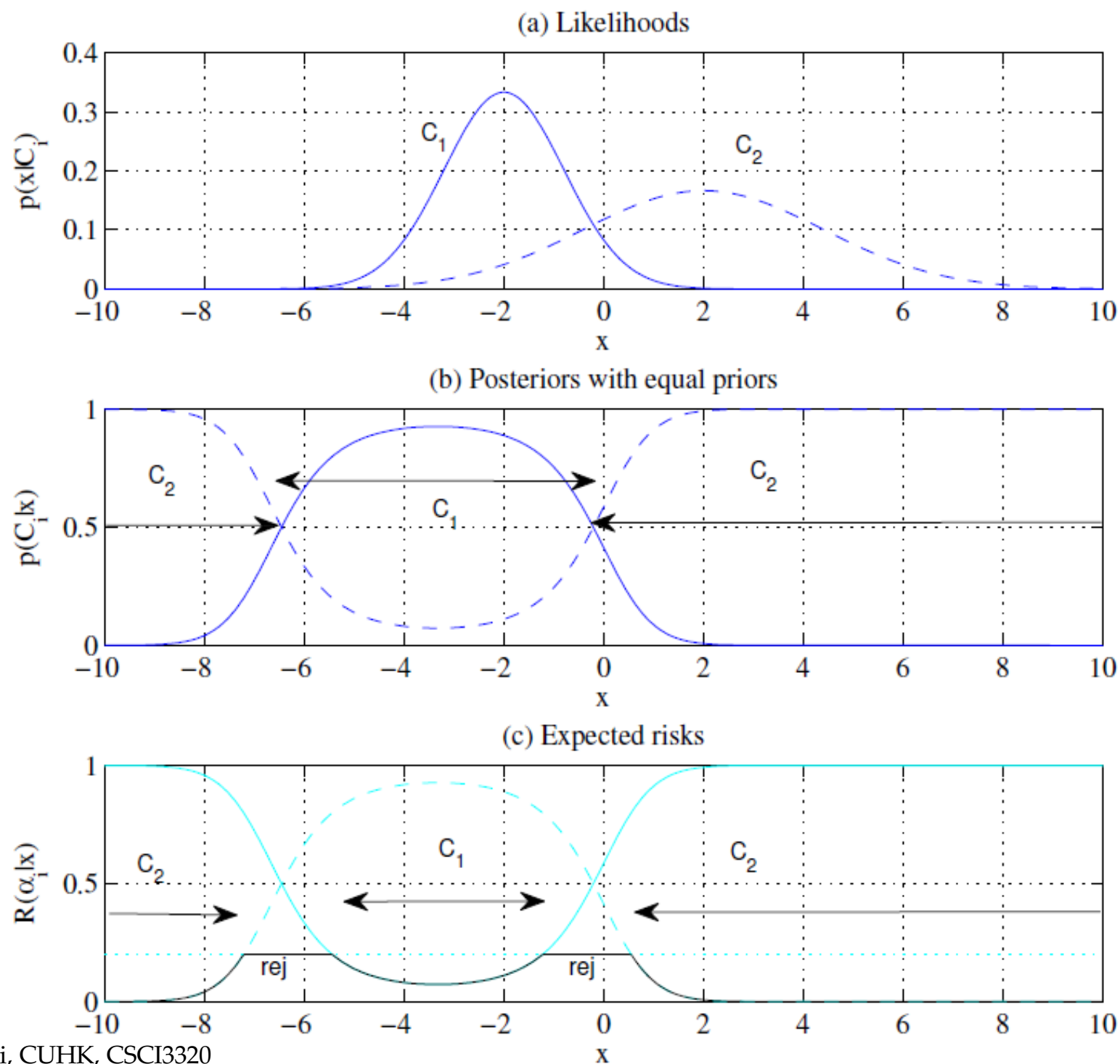
When variance (s_i^2) are different



Two boundaries.

Homework !!!





More comment:

- If we have prior information about the means, we can use a Bayesian estimate of $p(x|C_i)$ with prior on μ_i . We will cover this in later chapter
- When x is continuous, don't assume it is Gaussian
- Use the data and test whether it is Gaussian
 - ▣ Via plotting to visually check (**remember QQ plot?**)
 - ▣ Use ***normality test*** in statistics
- Summary of current “***likelihood approach***” in classification
 - ▣ **Use data to estimate densities separately**
 - ▣ **Calculate posterior densities with Bayes' rules**
 - ▣ **Derive discriminant**

Evaluating an Estimator: Bias and Variance

Unknown parameter θ

Estimator $d_i = h(X_i)$ on sample X_i

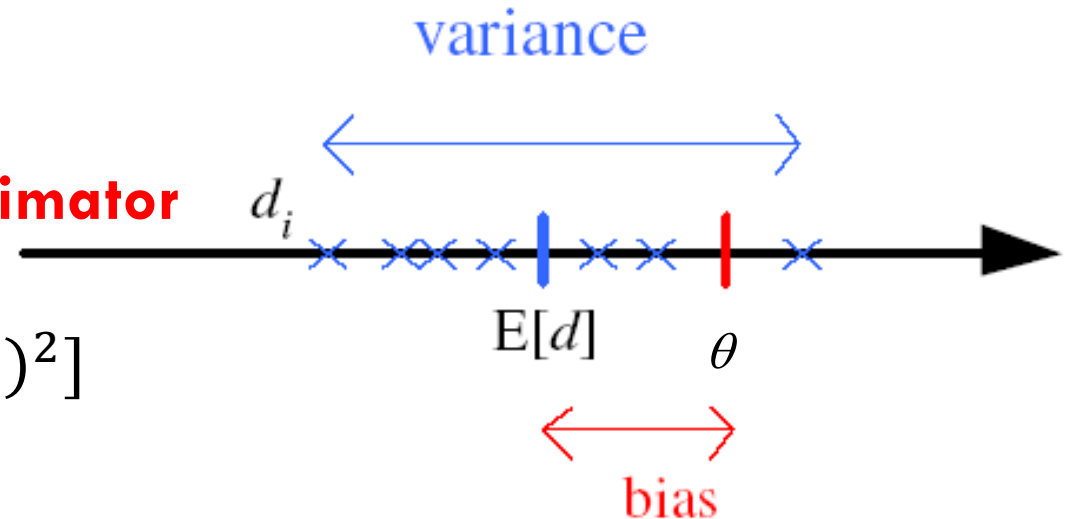
Bias: $b_{\theta(d)} = E[d] - \theta$

when $b_{\theta(d)} = 0$, **unbiased estimator**

Variance of RV d : $E[(d - E[d])^2]$

Mean square error:

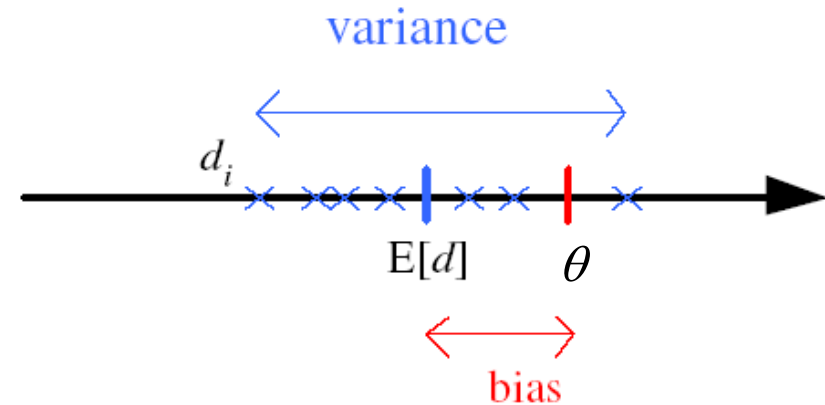
$$\begin{aligned} r(d, \theta) &= E[(d - \theta)^2] \\ &= (E[d] - \theta)^2 + E[(d - E[d])^2] \\ &= \text{Bias}^2 + \text{Variance} \end{aligned}$$



Evaluating an Estimator: Bias and Variance

Mean square error:

$$\begin{aligned} r(d, \theta) &= E[(d - \theta)^2] \\ &= (E[d] - \theta)^2 + E[(d - E[d])^2] \\ &= \text{Bias}^2 + \text{Variance} \end{aligned}$$



$$\begin{aligned} r(d, \theta) &= E[(d - \theta)^2] \\ &= E[(d - E[d] + E[d] - \theta)^2] \\ &= E[(d - E[d])^2 + (E[d] - \theta)^2 + 2(E[d] - \theta)(d - E[d])] \\ &= E[(d - E[d])^2] + E[(E[d] - \theta)^2] + 2E[(E[d] - \theta)(d - E[d])] \\ &= E[(d - E[d])^2] + (E[d] - \theta)^2 + 2(E[d] - \theta)E[d - E[d]] \\ &= \underbrace{E[(d - E[d])^2]}_{\text{variance}} + \underbrace{(E[d] - \theta)^2}_{\text{bias}^2} = 0 \end{aligned}$$

Evaluating an Estimator: Unbiased Estimator

x^t drawn from some density with mean μ

sample average, m , is an unbiased estimator

$$E[m] = E\left[\frac{\sum_t x^t}{N}\right] = \frac{1}{N} \sum_t E[x^t] = \frac{N\mu}{N} = \mu \quad \text{meaning}$$

m is also a consistent estimator, that is, $\text{Var}(m) \rightarrow 0$ as $N \rightarrow \infty$.

$$\text{Var}(m) = \text{Var}\left(\frac{\sum_t x^t}{N}\right) = \frac{1}{N^2} \sum_t \text{Var}(x^t) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N} \quad \text{meaning}$$

Evaluating an Estimator: Unbiased Estimator

s^2 , the MLE of σ^2 :

$$s^2 = \frac{\sum_t (x^t - m)^2}{N} = \frac{\sum_t (x^t)^2 - Nm^2}{N}$$
$$E[s^2] = \frac{\sum_t E[(x^t)^2] - N \cdot E[m^2]}{N} \quad \text{show in class}$$

Given that $\text{Var}(X) = E[X^2] - E[X]^2$, we get $E[X^2] = \text{Var}(X) + E[X]^2$,
show in class

$$E[(x^t)^2] = \sigma^2 + \mu^2 \text{ and } E[m^2] = \sigma^2/N + \mu^2$$

plugging these in, we get

$$E[s^2] = \frac{N(\sigma^2 + \mu^2) - N(\sigma^2/N + \mu^2)}{N} = \left(\frac{N-1}{N}\right) \sigma^2 \neq \sigma^2$$

s^2 is a biased estimator of σ^2

But an asymptotically unbiased estimator

Unbiased estimator: $(N/(N-1))s^2$

Bayes' Estimator (my notes)

- Sometimes, we know the possible value range of θ
- Treat θ as a random variable with prior $p(\theta)$
- Bayes' rule: $p(\theta|\mathcal{X}) = \frac{p(\mathcal{X}|\theta)p(\theta)}{p(\mathcal{X})}$
- To estimate density at x , we have

$$p(x|\mathcal{X}) = \int p(x|\theta')f(\theta')d\theta'$$


$$p(x|\mathcal{X}) = \sum_{\theta'} p(x|\theta')P(\theta')d\theta'$$

Bayes' Estimator (my note)

□ **Maximum a Posteriori (MAP):** $\theta_{MAP} = \arg \max_{\theta} p(\theta|\mathcal{X})$

Thus, we have $p(x|\mathcal{X}) = p(x|\theta_{MAP})$

□ If no prior values of θ , prior density $p(\theta)$ is flat, MAP estimate is equivalent to maximum likelihood estimate

□ **Maximum Likelihood (ML):** $\theta_{ML} = \arg \max_{\theta} p(\mathcal{X}|\theta)$

□ **Bayes' estimator** (or expected value of the posterior probability):

$$E[\theta|\mathcal{X}] = \int \theta p(\theta|\mathcal{X}) d\theta$$

Bayes' Estimator: Example

$x^t \sim \mathcal{N}(\theta, \sigma^2)$ and $\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$, μ_0 , σ_0^2 , and σ^2 are known

$$p(\mathcal{X}|\theta) = \frac{1}{(2\pi)^{N/2} \sigma^N} \exp \left[-\frac{\sum_t (x^t - \theta)^2}{2\sigma^2} \right]$$
$$p(\theta) = \frac{1}{\sqrt{2\pi} \sigma_0} \exp \left[-\frac{(\theta - \mu_0)^2}{2\sigma_0^2} \right]$$

reduce the posterior density to a single pt

It can be shown that $p(\theta|\mathcal{X})$ is normal with

$$E[\theta|\mathcal{X}] = \frac{N/\sigma^2}{N/\sigma^2 + 1/\sigma_0^2} m + \frac{1/\sigma_0^2}{N/\sigma^2 + 1/\sigma_0^2} \mu_0$$

Thus the Bayes' estimator is a weighted average of the prior mean μ_0 and the sample mean m , with weights being inversely proportional to their variances. As the sample size N increases, the Bayes' estimator gets closer to the sample average, using more the information provided by the sample. When σ_0^2 is small, that is, when we have little prior uncertainty regarding the correct value of θ , or when N is small, our prior guess μ_0 has a higher effect.

Example of Bayes' Estimator

Let $(X_1, X_2, \dots, X_n)$ be the random sample of a Bernoulli R.V. X with

$$P(x) = p^x(1-p)^{1-x} \quad x = 0, 1.$$

p is unknown but it is uniformly distributed on $(0, 1)$, or

$$f(p) = 1 \quad 0 < p < 1.$$

The posterior pdf of p is: $f(p|x_1, \dots, x_n) = \frac{f(x_1, \dots, x_n, p)}{P(x_1, \dots, x_n)}$

$$\begin{aligned} f(x_1, \dots, x_n, p) &= f(x_1, \dots, x_n|p)f(p) \\ &= p^{\sum_{i=1}^n x_i} (1-p)^{n-\sum_{i=1}^n x_i} = p^m (1-p)^{n-m} \times 1 \end{aligned}$$

where $m = \sum_{i=1}^n x_i$

Example of Bayes' Estimator

$$P(x_1, \dots, x_n) = \int_0^1 f(x_1, \dots, x_n, p) dp = \int_0^1 p^m (1-p)^{n-m} dp$$

From calculus, we have: $\int_0^1 p^m (1-p)^k dp = \frac{m!k!}{(m+k+1)!}$

$$\text{So: } f(p|x_1, \dots, x_n) = \frac{f(x_1, \dots, x_n, p)}{P(x_1, \dots, x_n)} = \frac{(n+1)! p^m (1-p)^{n-m}}{m!(n-m)!}$$

Example of Bayes' Estimator

Bayes' estimator $E[p|\mathcal{X}]$:

$$\begin{aligned} E[p|\mathcal{X}] &= \int_0^1 p f(p|x_1, \dots, x_n) dp \\ &= \frac{(n+1)!}{m!(n-m)!} \int_0^1 p^{(m+1)} (1-p)^{n-m} dp \\ &= \frac{(n+1)!}{m!(n-m)!} \frac{(m+1)!(n-m)!}{(n+2)!} = \frac{m+1}{n+2} \\ &= \frac{1}{n+2} \left(\sum_{i=1}^n x_i + 1 \right) \end{aligned}$$

Bayes' estimator p_B is $\frac{1}{n+2} (\sum_{i=1}^n x_i + 1)$

Regression

r : dependent variable **unknown function $f()$**

x : independent variable

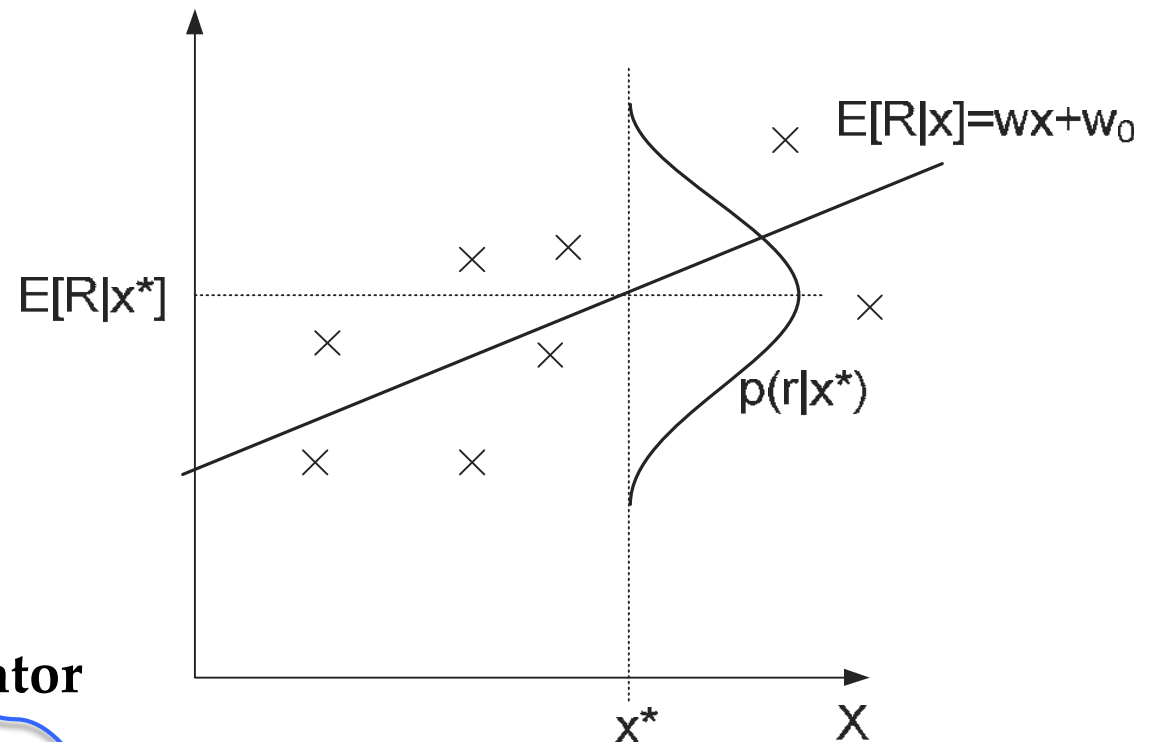
$$r = f(x) + \varepsilon$$

noise

estimator: $g(x|\theta)$

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

$$p(r|x) \sim \mathcal{N}(g(x|\theta), \sigma^2)$$



log maximum likelihood estimator

$$\mathcal{L}(\theta|\mathcal{X}) = \log \prod_{t=1}^N p(x^t, r^t)$$

$$= \log \prod_{t=1}^N p(r^t|x^t) + \log \prod_{t=1}^N p(x^t)$$

can be ignored

Regression: From LogL to Error

$$\begin{aligned}\mathcal{L}(\theta|\mathcal{X}) &= \log \prod_{t=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{[r^t - g(x^t|\theta)]^2}{2\sigma^2} \right] \\ &= \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^N \exp \left[-\frac{1}{2\sigma^2} \sum_{t=1}^N [r^t - g(x^t|\theta)]^2 \right] \\ &= -N \log(\sqrt{2\pi}\sigma) - \frac{1}{2\sigma^2} \sum_{t=1}^N [r^t - g(x^t|\theta)]^2\end{aligned}$$

First term is not a function of θ , maximize the above term is equivalent to **minimize** the following **error expression** (E):

$$E(\theta|\mathcal{X}) = \frac{1}{2} \sum_{t=1}^N [r^t - g(x^t|\theta)]^2 \quad \theta \text{ that minimize this function is the } \textit{least square estimates}$$

Example: Linear Regression

$$g(x^t | w_1, w_0) = w_1 x^t + w_0$$

Using the error function in previous slide, taking derivatives for two unknowns, we have:

$$\sum_t r^t = N w_0 + w_1 \sum_t x^t$$

$$\sum_t r^t x^t = w_0 \sum_t x_t + w_1 \sum_t (x^t)^2$$

vector-matrix form as $\mathbf{A}\mathbf{w} = \mathbf{y}$

$$\mathbf{A} = \begin{bmatrix} N & \sum_t x^t \\ \sum_t x^t & \sum_t (x^t)^2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \sum_t r^t \\ \sum_t r^t x^t \end{bmatrix}$$

2 x 2 matrix *2 x 1 vector* *2 x 1 vector*

solved as $\mathbf{w} = \mathbf{A}^{-1}\mathbf{y}$.

Example: Polynomial Regression of **order k**

$$g(x^t | w_k, \dots, w_2, w_1, w_0) = w_k (x^t)^k + \dots + w_2 (x^t)^2 + w_1 x^t + w_0$$

Taking $(k+1)$ derivatives of E , we have $(k+1)$ equations :

$\mathbf{A}\mathbf{w} = \mathbf{y}$ where

$$\mathbf{A} = \begin{bmatrix} N & \sum_t x^t & \sum_t (x^t)^2 & \dots & \sum_t (x^t)^k \\ \sum_t x^t & \sum_t (x^t)^2 & \sum_t (x^t)^3 & \dots & \sum_t (x^t)^{k+1} \\ \vdots & & & & \\ \sum_t (x^t)^k & \sum_t (x^t)^{k+1} & \sum_t (x^t)^{k+2} & \dots & \sum_t (x^t)^{2k} \end{bmatrix}$$
$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_k \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \sum_t r^t \\ \sum_t r^t x^t \\ \sum_t r^t (x^t)^2 \\ \vdots \\ \sum_t r^t (x^t)^k \end{bmatrix}$$

Example: Polynomial Regression of **order k**

We can write $\mathbf{A} = \mathbf{D}^T \mathbf{D}$ and $\mathbf{y} = \mathbf{D}^T \mathbf{r}$ where

$$\mathbf{D} = \begin{bmatrix} 1 & x^1 & (x^1)^2 & \dots & (x^1)^k \\ 1 & x^2 & (x^2)^2 & \dots & (x^2)^k \\ \vdots & & & & \\ 1 & x^N & (x^N)^2 & \dots & (x^N)^k \end{bmatrix}, \mathbf{r} = \begin{bmatrix} r^1 \\ r^2 \\ \vdots \\ r^N \end{bmatrix}$$

solve for the parameters as

$$\mathbf{w} = (\mathbf{D}^T \mathbf{D})^{-1} \mathbf{D}^T \mathbf{r}$$

Other Error Measures

□ **Square Error:**

$$E(\theta|\mathcal{X}) = \frac{\sum_{t=1}^N [\mathbf{r}^t - g(\mathbf{x}^t|\theta)]^2}{2}$$

□ **Relative Square Error:** $E(\theta|\mathcal{X}) = \frac{\sum_{t=1}^N [\mathbf{r}^t - g(\mathbf{x}^t|\theta)]^2}{\sum_{t=1}^N (\mathbf{r}^t - \bar{\mathbf{r}})^2}$

□ **Absolute Error:** $E(\theta|\mathcal{X}) = \sum_{t=1}^N |\mathbf{r}^t - g(\mathbf{x}^t|\theta)|$

□ **ϵ -sensitive Error:**

$$E(\theta|\mathcal{X}) = \sum_{t=1}^N 1 (|\mathbf{r}^t - g(\mathbf{x}^t|\theta)| > \epsilon) (|\mathbf{r}^t - g(\mathbf{x}^t|\theta)| - \epsilon)$$

Bias and Variance

The expected square error at the point x is:

$$E[(r - g(x))^2 | x] = \underbrace{E[(r - E[r|x])^2 | x]}_{\substack{\text{noise} \\ \text{variance}}} + \underbrace{(E[r|x] - g(x))^2}_{\text{squared error} \\ \text{bias}^2}$$

The expected value (average over samples X , all of size N and drawn from the same joint density $p(r, x)$ is):

$$E_X[(E[r|x] - g(x))^2 | x] = \underbrace{(E[r|x] - E_X[g(x)])^2}_{\text{bias}^2} + \underbrace{E_X[(g(x) - E_X[g(x)])^2]}_{\text{variance}}$$

Illustrating Bias and Variance

- M samples $X_i = \{x_i^t, r_i^t\}, i = 1, \dots, M$ of **known function $f()$** with added noise. Use each data sets to fit estimator $g_i(x), i = 1, \dots, M$
- $E[g(x)]$ is estimated by average over M of $g_i()$

$$\bar{g}(x) = \frac{1}{M} \sum_{i=1}^M g_i(x)$$

$$\text{Bias}^2(g) = \frac{1}{N} \sum_t [\bar{g}(x^t) - f(x^t)]^2$$

$$\text{Variance}(g) = \frac{1}{NM} \sum_t \sum_i [g_i(x^t) - \bar{g}(x^t)]^2$$

Bias/Variance Dilemma (or tradeoff)

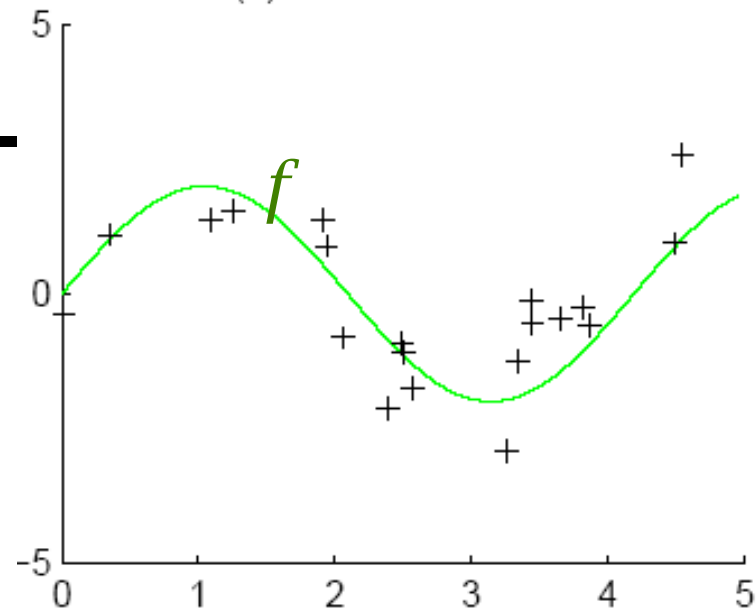
- If $g_i(x) = 2$: It has **no variance and high bias**
- If $g_i(x) = \sum_t \frac{r_i^t}{N}$: It has **lower bias, higher variance**
- As we increase complexity,
 bias decreases (a better fit to data) and
 variance increases (fit varies more with data)
- Bias/Variance dilemma: (Geman et al., 1992)

Illustration

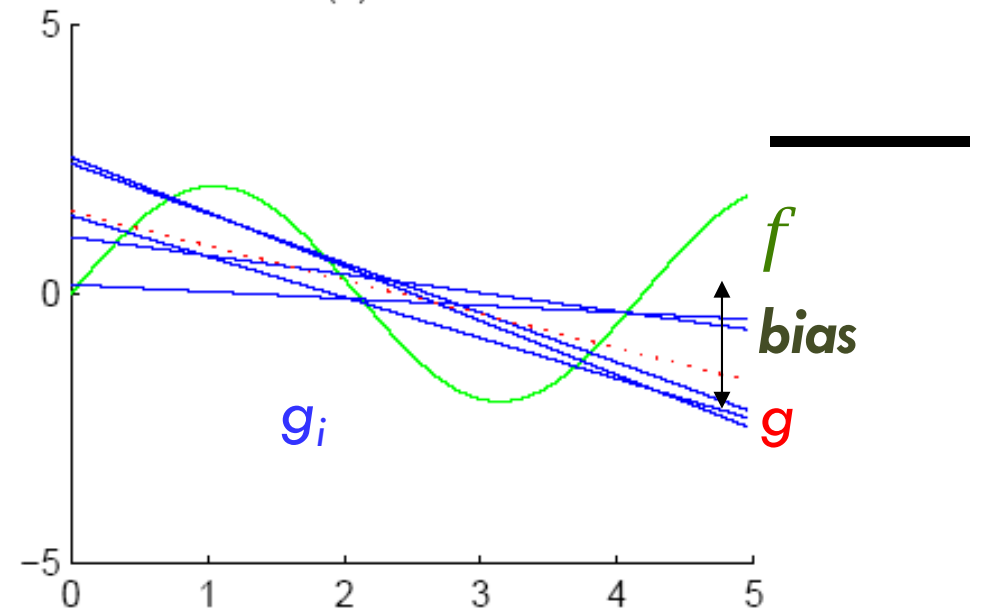
$$r = f(x) + \varepsilon$$

- **Function:** $f(x) = 2 \sin(1.5x)$
- **Noise:** $(\mathcal{N}(0, 1))$
- **Samples:** $M=5$ samples are taken, each contains $N=20$ points
- **Five polynomial fit:**
 $g_i(x), i=1, \dots, 5$ of order 1, 3, and 5

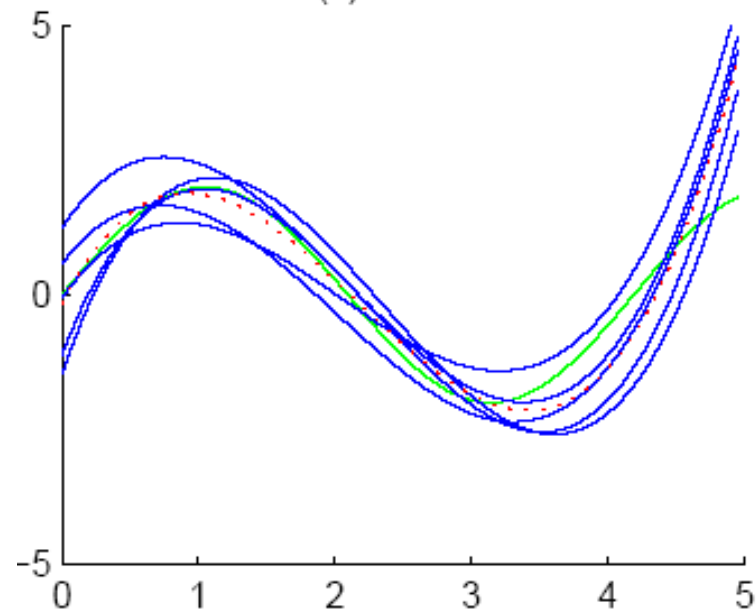
(a) Function and data



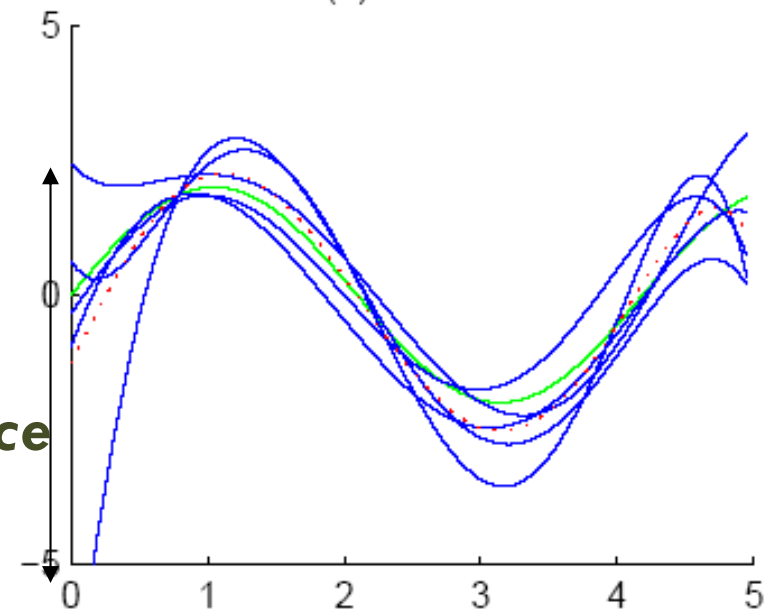
(b) Order 1



(c) Order 3

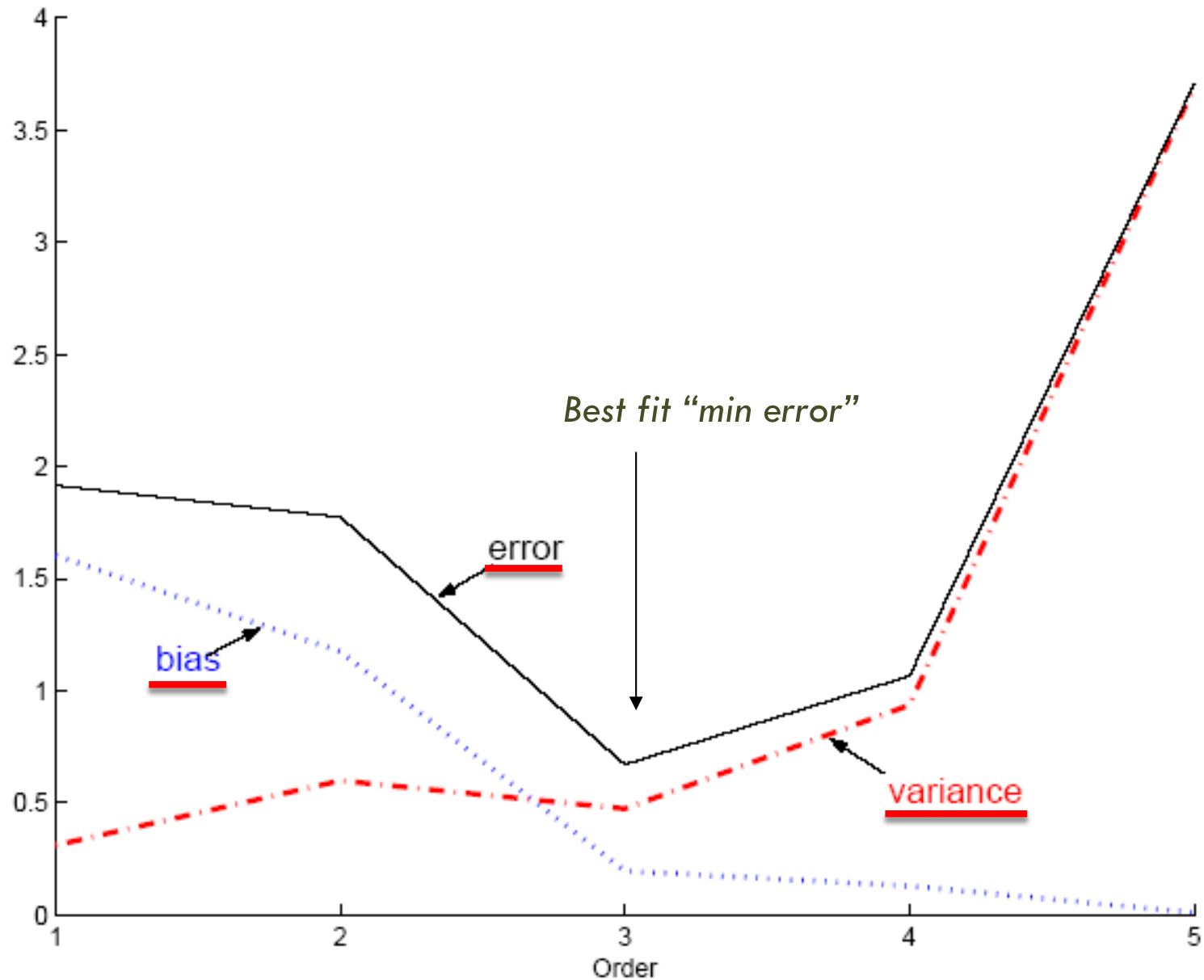


(d) Order 5



variance

Polynomial Regression ($M=100$)

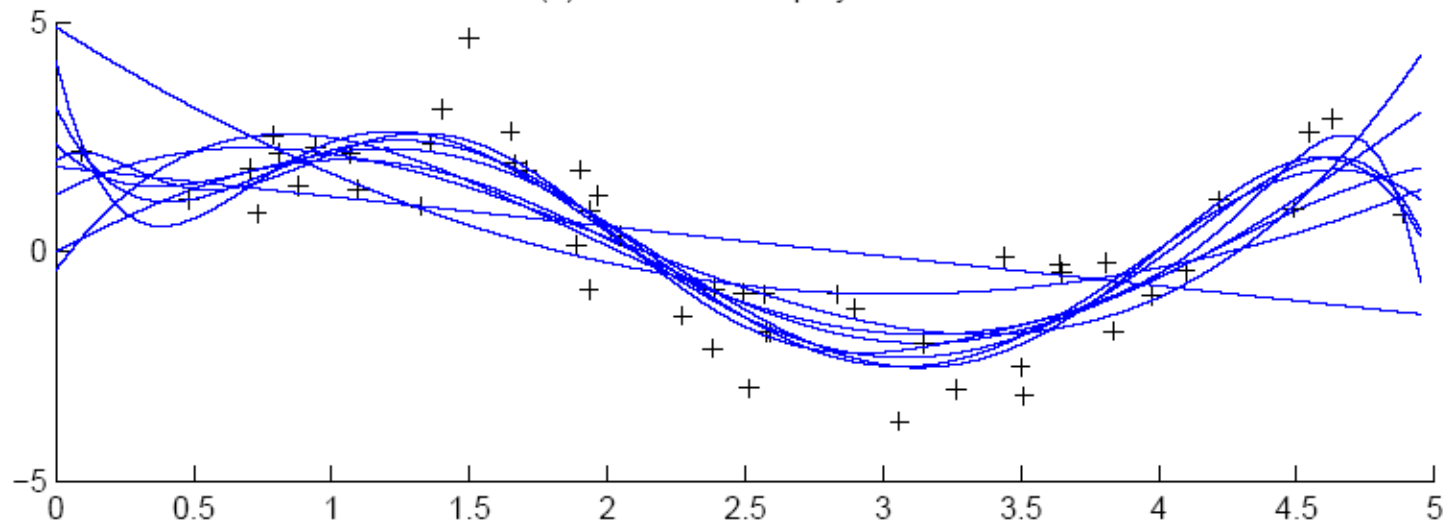


Underfitting vs. Overfitting

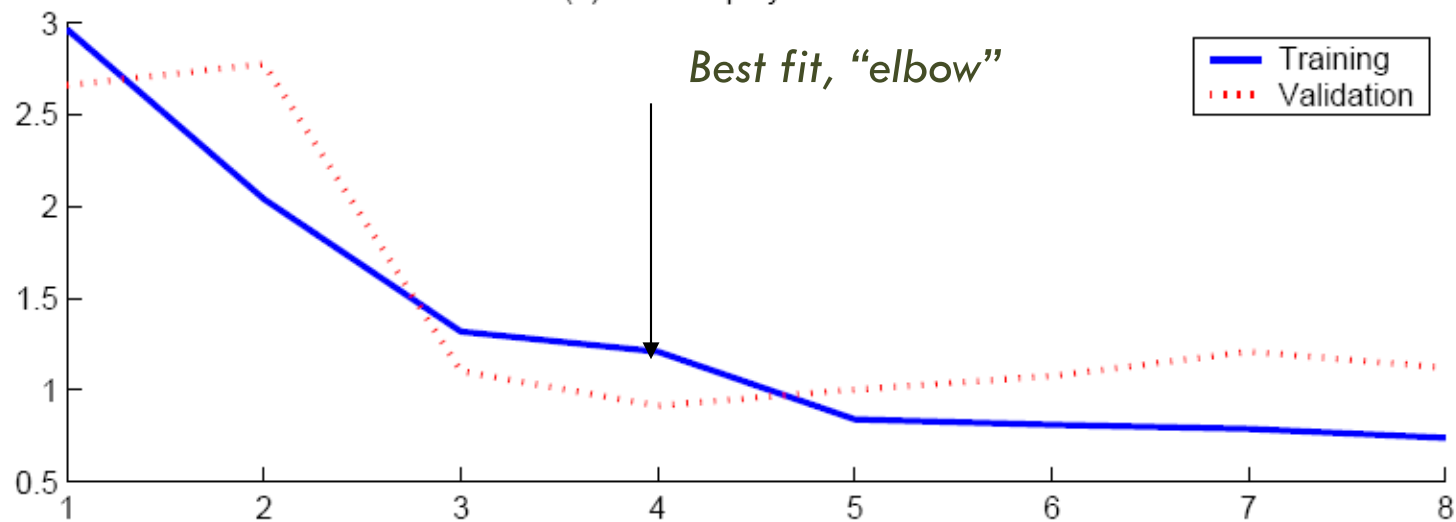
- High bias implies “*underfitting*”
- High variance implies “*overfitting*”
- Tradeoff via *cross-validation*

M=50

(a) Data and fitted polynomials



(b) Error vs polynomial order



Model Selection

- **Cross-validation:** Measure generalization accuracy by testing on data unused during training
- **Regularization:** Penalize complex models
 $E' = \text{error on data} + \lambda \text{ model complexity}$
Akaike's information criterion (AIC), Bayesian information criterion (BIC)
- **Minimum description length (MDL):** Kolmogorov complexity, shortest description of data
- **Structural risk minimization (SRM)**

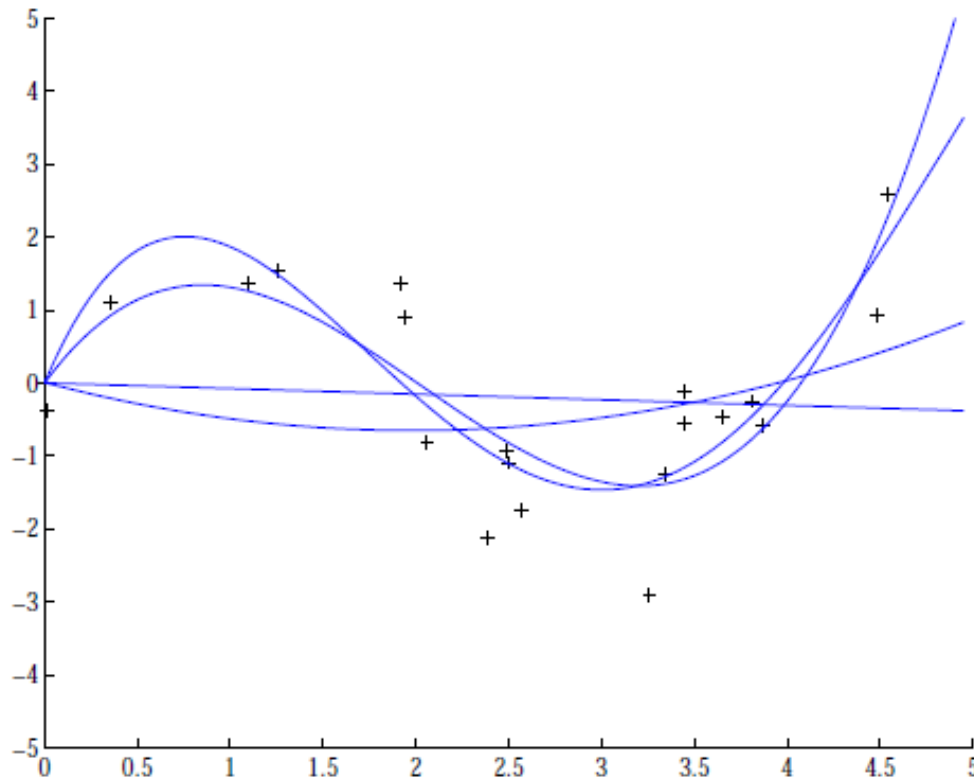
Bayesian Model Selection

- Prior on models, $p(\text{model})$

$$p(\text{model} | \text{data}) = \frac{p(\text{data} | \text{model}) p(\text{model})}{p(\text{data})}$$

- Regularization, when prior favors simpler models
- Bayes, MAP of the posterior, $p(\text{model} | \text{data})$
- Average over a number of models with high posterior (voting, ensembles: Chapter 17)

Regression example



Coefficients increase in magnitude as order increases:

1: [-0.0769, 0.0016]

2: [0.1682, -0.6657, 0.0080]

3: [0.4238, -2.5778, 3.4675, -0.0002]

4: [-0.1093, 1.4356, -5.5007, 6.0454, -0.0019]

Regularization (L2):
$$E = \sum_t [r^t - g(x^t | \mathbf{w})]^2 + \lambda \sum_i w_i^2$$

Conclusion

- Simple Bayesian (or Bayes' rule) has some shortcomings, in particular, when x is not in the database/training data. So we have problem getting the likelihood (or evidence)
- In parametric methods, we assume x follows some probability distribution. So if we can have an "accurate" estimate of the parameters of the distribution, we can use the probability density for the point x , this can overcome the problem in the Bayesian method.
- We discuss how to obtain the MLE for parameters in Bernoulli, multinomial, Gaussian probability distribution
- We discuss how to do regression using parametric method
- We discuss Unbiased estimation
- We discuss mean square error = $bias^2 + variance$
- We discuss their tradeoffs.